Sustainable Supply Chain Resilience in Cross-Border Contexts:

A Multi-Stakeholder Perspective on Risk Management and Circular Economy Integration

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Abstract

The increasing complexity of cross-border supply chains, coupled with the growing urgency of sustainability imperatives, demands a fundamental reconceptualization of supply chain resilience. This paper develops an integrated framework that synthesizes sustainable supply chain management (SSCM) with resilience theory and circular economy principles in the context of cross-border operations. Drawing on a multi-stakeholder case study methodology involving 12 multinational enterprises operating across Asia–Europe and Asia–North America trade corridors, we identify four critical capabilities for sustainable cross-border resilience: (1) multi-tier visibility and traceability, (2) adaptive circular sourcing, (3) collaborative risk governance across institutional boundaries, and (4) digital-enabled ecosystem orchestration. Our findings reveal that firms with higher levels of sustainable supply chain maturity are significantly better equipped to absorb disruptions while maintaining both economic and environmental performance. The study contributes to the supply chain management literature by extending resilience theory to incorporate sustainability dimensions in cross-border settings, and by providing a practical maturity model for managers navigating the twin challenges of sustainability and resilience in global supply networks.

Keywords: Sustainable supply chain management; Supply chain resilience; Cross-border supply chains; Circular economy; Risk governance; Multi-stakeholder collaboration

1 Introduction

Global supply chains are confronting an unprecedented confluence of disruptions — from geopolitical tensions and trade policy volatility to climate-induced extreme weather events and public health emergencies (**??**). Simultaneously, stakeholder pressure for environmental and social sustainability has moved from the periphery to the core of corporate strategy, reshaping how supply chains are designed, governed, and operated (**??**).

The intersection of these two imperatives — resilience and sustainability — is particularly acute in cross-border supply chains, where firms must navigate heterogeneous regulatory environments, diverse cultural norms, and complex multi-tier supplier networks that span multiple jurisdictions (**?**). Recent disruptions, including the COVID-19 pandemic, the blockage of the Suez Canal, and ongoing trade tensions between major economies, have exposed the fragility of globally dispersed, lean-optimized supply networks (**??**).

While the supply chain management literature has separately addressed resilience (**??**) and sustainability (**??**), there remains a significant gap in understanding how these two paradigms interact and can be mutually reinforcing in cross-border contexts. Existing frameworks tend to treat resilience and sustainability as parallel or even competing objectives, rather than as potentially synergistic dimensions of supply chain design (**??**).

This study addresses this gap by investigating the following research questions:

1. How can sustainable supply chain management (SSCM) practices enhance resilience in cross-border supply networks?
2. What are the critical capabilities that enable firms to simultaneously pursue sustainability and resilience objectives across national boundaries?
3. How do institutional differences between countries moderate the relationship between SSCM and resilience?

We adopt a multi-stakeholder case study approach involving 12 multinational enterprises (MNEs) operating across two major trade corridors: Asia–Europe and Asia–North America. This comparative design allows us to examine how institutional and cultural differences shape the sustainability–resilience nexus in cross-border supply chains.

Our study makes three principal contributions. First, we develop an integrated conceptual framework that identifies four critical capabilities for sustainable cross-border resilience. Second, we extend resilience theory by incorporating circular economy principles as resilience-building mechanisms. Third, we provide a practical maturity model that helps managers assess and develop their organizations' capacity for sustainable supply chain resilience.

The remainder of this paper is organized as follows. Section 2 reviews the literature on SSCM, supply chain resilience, and cross-border operations. Section 3 describes our research methodology. Section 4 presents the case study findings organized around the four capability dimensions. Section 5 synthesizes the findings into an integrated framework. Section 6 discusses theoretical and practical implications, and Section 7 concludes with limitations and future research directions.

2 Literature Review

2.1 Sustainable Supply Chain Management

Sustainable supply chain management (SSCM) has evolved from a niche concern into a central research domain within operations and supply chain management (**??**). SSCM is defined as "the strategic, transparent integration and achievement of an organization's social, environmental, and economic goals in the systemic coordination of key inter-organizational business processes for improving the long-term economic performance of the individual company and its supply chains" (**?**, p. 1700).

The SSCM literature has examined a wide range of practices, including green purchasing, sustainable supplier development, reverse logistics, and circular economy initiatives (**??**). More recently, scholars have emphasized the importance of extending sustainability efforts beyond the first tier of suppliers to the entire supply network (**??**).

A growing body of work explores the relationship between SSCM and firm performance.

Meta-analyses suggest that sustainable supply chain practices are positively associated with both environmental and economic performance, though the effects are moderated by contextual factors such as industry, region, and institutional environment (**??**).

2.2 Supply Chain Resilience

Supply chain resilience refers to "the adaptive capability of the supply chain to prepare for unexpected events, respond to disruptions, and recover from them by maintaining continuity of operations at desired levels of connectedness and control over structure and function" (**?**, p. 131). The resilience literature has identified a range of capabilities and strategies, including redundancy, flexibility, visibility, collaboration, and agility (**???**).

Recent research has shifted from a reactive view of resilience (bouncing back after disruptions) toward a more proactive, capability-based perspective (**??**). This evolution recognizes that resilience is not merely about recovery but about building the capacity to anticipate, adapt to, and even thrive amid change (**??**).

However, the resilience literature has been criticized for its limited attention to sustainability dimensions. Most resilience frameworks focus exclusively on operational and economic performance, overlooking the environmental and social implications of resilience-building strategies (**??**).

2.3 Cross-Border Supply Chain Operations

Cross-border supply chains introduce additional layers of complexity that distinguish them from domestic operations. Institutional theory suggests that firms operating across borders must navigate different regulatory, normative, and cognitive institutional environments (**??**). These institutional differences affect everything from supplier selection and relationship management to logistics and compliance (**?**).

The "liability of foreignness" (**?**) manifests in supply chain contexts through challenges such as cultural misunderstandings, differences in business practices, and varying levels of institutional trust (**??**). These challenges are particularly salient for sustainability initiatives, which often require deep collaboration and information sharing across cultural and institutional boundaries (??).

Cross-border supply chains also face unique risks, including currency fluctuations, trade policy changes, customs delays, and geopolitical instability (??). The intersection of these risks with sustainability pressures creates a complex landscape for supply chain managers (?).

2.4 Integrating Sustainability and Resilience

Recent scholarship has begun to explore the nexus between sustainability and resilience in supply chains. ? proposed that green supply chain practices can enhance resilience by diversifying sourcing options and reducing dependence on environmentally risky suppliers. ? developed a conceptual model linking circular economy principles to resilience capabilities, arguing that circular strategies such as remanufacturing and recycling can serve as buffers against supply disruptions.

However, tensions between sustainability and resilience objectives have also been identified. For example, the efficiency-oriented logic of lean supply chains — often associated with both cost reduction and environmental benefits through waste elimination — can conflict with the redundancy and slack resources needed for resilience (?). Similarly, the localization of supply chains, often advocated on sustainability grounds, may reduce the flexibility benefits of global sourcing (?).

Table 1 summarizes key studies at the sustainability–resilience interface in supply chain research.

As Table 1 illustrates, existing research has largely focused on single-country contexts or conceptual modeling. Empirical research on the sustainability–resilience nexus in cross-border supply chains remains scarce, motivating the present study.

3 Research Methodology

3.1 Research Design

We adopted a multi-stakeholder, comparative case study design (??) to investigate the sustainability–resilience nexus in cross-border supply chains. The case study method is particularly appropriate for exploring complex, context-dependent phenomena where existing theory is underdevel-

Table 1: *Selected Studies at the Sustainability–Resilience Interface*

Study	Focus	Key Finding	Context
?	Green SC practices and resilience	Green practices enhance disruption absorption	Single-country
?	Circular economy and resilience	Circularity creates buffers against disruption	Conceptual
?	SSCM and performance	Sustainability moderates disruption impact	Multi-industry
?	Resilient-sustainable network design	Trade-offs between cost and sustainability	Mathematical modeling

oped (?).

3.2 Case Selection

Twelve multinational enterprises were selected through theoretical sampling (?) to ensure variation across key dimensions: geographic scope, industry sector, and SSCM maturity. The cases span two major trade corridors:

1. **Asia–Europe Corridor** (7 cases): Firms with supply networks connecting manufacturing bases in China, Vietnam, and India with European markets, primarily in the automotive, electronics, and textile sectors.

2. **Asia–North America Corridor** (5 cases): Firms with supply networks connecting Asia-Pacific production hubs with North American markets, spanning consumer goods, industrial equipment, and pharmaceutical sectors.

Table 2 provides an overview of the case organizations.

3.3 Data Collection

Data were collected between January 2024 and December 2025 through multiple sources to enable triangulation (?):

Table 2: Overview of Case Organizations

Firm	Sector	Employees	Corridor	SSCM Maturity
AutoCorp	Automotive	25,000+	Asia–Europe	High
ElectroTech	Electronics	12,000	Asia–Europe	Medium
TextileGroup	Textiles	8,000	Asia–Europe	Low
PharmaMed	Pharmaceuticals	15,000	Asia–NA	High
ConsumerGoods	Consumer Goods	30,000+	Asia–NA	Medium
	··· (6 additional cases)			

- **Semi-structured interviews**: 96 interviews were conducted with supply chain directors, sustainability managers, procurement heads, and key supplier representatives (8 per case firm). Each interview lasted 45–90 minutes.

- **Document analysis**: Internal documents including sustainability reports, supplier codes of conduct, risk assessment frameworks, and disruption response plans.

- **Site visits**: Researchers conducted 24 site visits to manufacturing facilities, distribution centers, and supplier sites across six countries.

- **Archival data**: Publicly available data on disruptions, trade flows, and regulatory changes affecting the case firms' supply chains.

3.4 Data Analysis

We followed the Gioia methodology for qualitative data analysis (**?**), proceeding through three stages: (1) first-order analysis identifying informant-centric concepts, (2) second-order analysis developing researcher-centric themes, and (3) aggregation into theoretical dimensions. NVivo 14 was used for data management and coding. Cross-case analysis employed pattern-matching logic (**?**) to identify commonalities and differences across cases.

4 Findings

Our analysis identified four critical capabilities that enable firms to simultaneously pursue sustainability and resilience objectives in cross-border supply chains. We present each capability below, illustrated with evidence from the case studies.

4.1 Capability 1: Multi-Tier Visibility and Traceability

The ability to see beyond first-tier suppliers emerged as a foundational capability for sustainable cross-border resilience. Firms with high SSCM maturity had invested in digital traceability systems that provided real-time visibility into the environmental and social performance of sub-tier suppliers.

As the Supply Chain Director of AutoCorp explained:

> *"During the 2024 semiconductor shortage, our upstream visibility — right down to the raw material level — was what saved us. We knew exactly which sub-tier suppliers were at risk and could redirect orders before production lines stopped. That same visibility also ensures our conflict minerals compliance across 14 countries."*

Firms with lower visibility maturity relied on manual supplier surveys and third-party audits, which proved inadequate for rapid disruption response. Table 3 summarizes the relationship between visibility maturity and resilience outcomes.

Table 3: *Multi-Tier Visibility and Resilience Outcomes*

	Visibility Maturity		
Outcome	**Low**	**Medium**	**High**
Disruption detection time	12–48 hours	4–12 hours	<4 hours
Recovery time (days)	14–30	7–14	3–7
Supplier compliance rate	62%	78%	94%

4.2 Capability 2: Adaptive Circular Sourcing

Circular economy practices — including closed-loop supply chains, remanufacturing, and recycled material sourcing — served dual sustainability and resilience functions. Firms with established circular sourcing networks were able to pivot to secondary material sources when primary supply was disrupted.

PharmaMed's experience during the 2024 trade restrictions illustrated this dynamic:

> *"When our primary solvent supplier in [Country X] was sanctioned, our investment in chemical recovery and recycling systems meant we could maintain 70% of production capacity using recovered solvents, while we qualified an alternative supplier. Without circularity, we would have faced a complete shutdown."*

Cross-case analysis revealed three circular sourcing strategies associated with enhanced resilience: (1) closed-loop material recovery within the firm's own operations, (2) industrial symbiosis networks with co-located firms, and (3) certified secondary material markets.

4.3 Capability 3: Collaborative Risk Governance

Cross-border supply chains operate across institutional boundaries, making collaborative risk governance essential. Our analysis identified three governance mechanisms that were particularly effective:

Multi-stakeholder risk councils. Six case firms had established formal risk councils comprising representatives from procurement, sustainability, legal, and key supplier organizations. These councils met quarterly to assess emerging risks and coordinate response strategies.

Institutional bridging. Successful firms employed "institutional bridging" strategies — deploying personnel with deep understanding of both home and host country institutional environments to facilitate communication and build trust across borders.

Contractual-relational hybrid governance. Following **?**, we observed that firms combined formal contractual mechanisms (e.g., sustainability clauses, disruption recovery SLAs) with relational mechanisms (e.g., joint problem-solving, information sharing) to govern cross-border supplier relationships. The balance shifted toward relational governance as institutional distance increased.

Table 4 maps governance mechanisms to institutional contexts.

Table 4: *Governance Mechanisms Across Institutional Contexts*

Institutional Distance	**Dominant Mechanism**	**Secondary Mechanism**	**Example**
Low (e.g., EU–EU)	Contractual	Relational	Detailed SLAs
Medium (e.g., EU–China)	Hybrid	Relational	Framework agreements
High (e.g., EU–SE Asia)	Relational	Contractual	Joint risk councils

4.4 Capability 4: Digital-Enabled Ecosystem Orchestration

The most advanced firms in our sample had evolved from managing dyadic supplier relationships to orchestrating multi-stakeholder ecosystems. Digital platforms served as the backbone for this orchestration, enabling:

- **Real-time risk sensing**: AI-powered monitoring of geopolitical, environmental, and supplier financial risk indicators across the supply network.
- **Collaborative planning**: Cloud-based platforms for joint demand forecasting, capacity planning, and inventory optimization with key suppliers.
- **Sustainability analytics**: Automated tracking of carbon footprint, water usage, and social compliance metrics across all tiers.

ElectroTech's Chief Procurement Officer described their ecosystem approach:

> *"We've moved from a hub-and-spoke model to a true ecosystem. Our digital platform connects 340 suppliers across 18 countries. When Typhoon Yagi hit Vietnam in 2024, the platform automatically identified affected suppliers, calculated impact on our production schedule, and suggested alternative sourcing options — all within two hours. Five years ago, that same process would have taken two weeks."*

5 Integrated Framework: The SCORE Model

Synthesizing our findings, we propose the **SCORE Model** — Sustainable Cross-border Operational Resilience Ecosystem — an integrated framework comprising five dimensions (Figure 1):

1. **Sensing**: Multi-tier visibility and early warning systems enabled by digital technologies.
2. **Circularity**: Closed-loop material flows and adaptive sourcing that provide redundancy without sacrificing sustainability.
3. **Orchestration**: Ecosystem-level coordination across institutional boundaries using digital platforms.
4. **Responsiveness**: Agile decision-making and rapid resource reconfiguration capabilities.
5. **Embeddedness**: Deep integration of sustainability values into organizational culture and supplier relationships.

[Insert Figure 1: The SCORE Framework]

Five dimensions: Sensing → Circularity → Orchestration → Responsiveness → Embeddedness
with cross-border institutional context as a moderating layer

Figure 1: *The SCORE Framework for Sustainable Cross-Border Resilience*

The framework emphasizes that these five dimensions are mutually reinforcing rather than independent. For example, sensing capabilities (dimension 1) enable more effective circular sourcing (dimension 2) by identifying opportunities for material recovery and reuse. Similarly, ecosystem orchestration (dimension 3) depends on the trust and shared values built through cultural embeddedness (dimension 5).

Table 5 presents a maturity model that helps firms assess their current position and identify development priorities for each SCORE dimension.

Table 5: *SCORE Maturity Model*

Dimension	Level 1: Reactive	Level 2: Proactive	Level 3: Regenerative
Sensing	Manual reporting, delayed detection	Automated monitoring, Tier-2 visibility	AI-driven, full-network transparency
Circularity	Basic recycling, waste reduction	Closed-loop systems, secondary markets	Industrial symbiosis, regenerative design
Orchestration	Dyadic relationships	Multi-tier coordination	Ecosystem co-creation
Responsiveness	Ad-hoc crisis response	Pre-defined contingency plans	Autonomous adaptation
Embeddedness	Compliance-driven	Values-aligned	Purpose-driven culture

6 Discussion

6.1 Theoretical Implications

This study makes three principal theoretical contributions to the supply chain management literature.

First, we extend resilience theory by demonstrating that sustainability practices are not merely compatible with resilience — they are enabling mechanisms. Specifically, circular economy practices create structural redundancy (multiple material sources) without the economic penalties typically associated with redundancy strategies. This finding challenges the prevailing assumption in the resilience literature that redundancy and efficiency are inherent trade-offs (**??**).

Second, we contribute to institutional theory in supply chain contexts by showing how institutional distance moderates the relationship between SSCM and resilience. Our finding that relational governance becomes more important as institutional distance increases extends **?**'s work on governance complementarity to the cross-border domain.

Third, we advance the SSCM literature by specifying the mechanisms through which sustainability investments generate resilience benefits. Rather than treating sustainability as a constraint on supply chain operations (**?**), our framework positions it as a strategic capability that enhances the adaptive capacity of cross-border supply networks.

6.2 Practical Implications

For supply chain managers, our findings offer several actionable insights:

1. **Invest in multi-tier visibility as a dual-purpose capability**. Digital traceability systems that serve sustainability compliance objectives (e.g., conflict minerals reporting, carbon accounting) also provide the visibility needed for rapid disruption detection and response.

2. **Develop circular sourcing as a resilience strategy**. Circular economy initiatives should be evaluated not only for their environmental benefits but also for their potential to create alternative material sources that buffer against supply disruptions.

3. **Adopt ecosystem-level governance for cross-border resilience**. Moving beyond dyadic supplier management to multi-stakeholder ecosystem orchestration enables more comprehensive risk assessment and coordinated response across institutional boundaries.

4. **Use the SCORE maturity model** (Table 5) to benchmark current capabilities and prioritize development investments across the five dimensions.

7 Conclusion

This study investigated how sustainable supply chain management practices can enhance resilience in cross-border supply networks. Through a multi-stakeholder case study of 12 multinational enterprises, we identified four critical capabilities — multi-tier visibility, adaptive circular sourcing, collaborative risk governance, and digital-enabled ecosystem orchestration — and synthesized them into the SCORE framework.

Our findings suggest that sustainability and resilience, far from being competing objectives, can be mutually reinforcing in cross-border contexts. Firms that have invested in SSCM capabilities are better positioned to detect, respond to, and recover from supply chain disruptions while maintaining environmental and social performance.

7.1 Limitations and Future Research

Several limitations should be acknowledged. First, our case selection focused on two specific trade corridors (Asia–Europe and Asia–North America); future research should examine whether our findings generalize to other corridors such as Africa–Asia or Latin America–North America. Second, our study captured a specific temporal window (2024–2025); longitudinal research is needed to understand how the sustainability–resilience relationship evolves over time. Third, while our qualitative design provides rich insights into mechanisms and processes, quantitative testing of the SCORE framework's propositions would strengthen its generalizability.

Future research could also explore the role of emerging technologies — particularly blockchain for supply chain transparency and artificial intelligence for predictive risk management — in enabling sustainable cross-border resilience. Additionally, the human and organizational dimensions of the sustainability–resilience nexus, including the role of leadership, culture, and change management, warrant deeper investigation.